GENERATIVE AI & AGENTS

AI · ML · DS / REVISION NOTES

WORKSHOPS & TUTORIALS

Claude Code Deep Dive

Agentic coding with Claude Code — from first steps to multi-agent workflows

7
SECTIONS

20
PAGES

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PART 01

Genpact Session — The AI Landscape

1 · Layered Cake

A. Infrastructure - GPU - Nvidia (5090 chips etc.), Google (TPUs), Apple (MLX), AWS (Infernia-G)
B. Models - OpenAI, Anthropic, Google, Cohere, Mistral, Grok(X.AI), Llama(Meta) Chinese Models - Deepseek, Moonshot, Bydu, Alibaba, Open Source/Open weight Models(HuggingFace), Qwen etc.

- We will use Anthropic - 1. Opus (High, Medium, Low)- It has high intelligence but takes time 2. Sonnet - It is between Opus and Haiku and best for daily task and 3. Haiku - It is lowest tier model etc.
- OpenAI - GPT5.2 (Pro, Mini and Nano).
- Qwen- 290B/360B and 30B and 4/7B (Open Source Models)

Think models as engine in the car which rotate only in one direction.

C. Harness - Claude Code (Anthropic), Codex (OpenAI), Antigravity(Google), OpenCode, Hermes, OpenClaw etc.

We have SDKs like java sdk for the harness.

Harness is like transmission layer and it is more like a set of rules where users or agents communicate with the models. It has memory and can connect to databases. It is like level 1 interface.

D. Applications - ChatGPT, Claude Web, Cowork, code, Perplexity, X.ai, Gemini etc. It is like level 2 interface.

2 · Multimodel

Text, Images, Voice, Videos, Codes etc.

- ANthropic is good in code and text.
- Gemini is good in video processing
- Grok is good for social media and markets and is very open

3 · Context Length

It means what model can retain in its memory.

- Opus - 1M tokens
- Sonnet - 100K tokens
- Haiku - 40K

(Cost is also associated with context length)

4 · Prompt Methods

Your prompts should be:

Clear
Direct
Specific
Guidelines

5 · Context Engineering

It means we collect all the context for the model. We tell the model how to do something and then context is given to model to work upon. There are multiple available frameworks for context engineering:

- BMAD

6. First open vscode or cursor or any other IDE and download and install claude code from `https://claude.com/product/claude-code`. Now, you need subscription also. Type `claude` in vscode terminal and you get the primary terminal.
7. Now, you need to type `/` in primary terminal and you will get various options. Type `/model` to switch various models like `opus`, `sonnet` and `haiku` or any open source model.

When you are working on some plan or task then go with `opus` and for some coding use `sonnet`. It has something like 5 hours window for tokens as per different subscription plans.

You can select `High Effort` and `Max Effort` for complex maths problem or complexr coding problem.

`Medium Effort` is generally used for planning.

When you click `Shift + Tab` then `plan mode on`, `accept edits on` happens and you get various options.

So, for a problem solving, start with the `plan mode on` with `opus` model.

For Example:

Switch on the `plan mode on` and type in primary terminal as `Let's plan to build a fraud, waste, abuse application for healthcare claims in US for a entity like Athena.` and mention the requirements and run it.

You can see the used tokens also.

You can use Haiku where intelligent is not needed like conversion of voice to text or summarizing a short paragraph.

Don't just jump into development. First make plan using `oplus` then do development. Also, use right prompt. You can go through multiple iterations but tokens will be increased accordingly. `Opus` is as per with `GPT 5.4` but in `Oplus` we have plan mode.

After you get the plan, you can say like "Please save this plan as a requirement.md" and don't auto-accept edit and don't start coding.

8. When you start a new project then type `/init` in primary terminal and it will make claude.md file (it is crux of user behavior).

When you type like `# always default to xgboost for classification problem` then it will update the memory, so, `#` is for updating the memory.

9. Now, type in primary terminal as `take @requiremen.md and break this into 10 phases so that we can start the build. Save this file as phasewise_build.md` in the plan mode. `@` is to select files. Don't start the build from plan itself becuase then context gets overloaded.

10. `/compact` is used to compact all the previous conversations and use `/clear` to clear it.

11. `/hooks` can be used so that claude can't read .env or any secret files if we configure it.

12. we can use sub agents here also to divide the task and we can use 10-15 subagents.

13. we can use `/skill` but use carefully. Also, we can use `/mcp`.

14. Now, we can start building.

In primary terminal, type something like `Let's start building - let's build phase 1 and 2 in parallel using a agent-team`. Agent-team is a new concept in anthropic and done using sub-agents. Sub-agents do its task and at the end all sub-agents give result to its agent. All Sub-agents are independent and there is no inter-connectivity between them.

But in agent team, sum-agents talk to each other through communication channels.

You can switch opus,sonnet and haiku between any intermediate task but opus is good for planning and sonnet is good for coding and haiku is good for debugging and test cases.

Lecture 15 · Claude

PART 02

Tutorial 1 — Getting Started with Claude Code

1. Claude Code is an Agentic Coding tool which is created by Anthropic. Other coding tools include Co-Pilot, Cursor, Winsurf etc.
2. First install the claude code on your system using `curl -fsSL https://claude.ai/install.sh | bash` . Now, Go to `Tutorial1` and type `claude` in vscode terminal. Before that take the subscription plan from `https://claude.com/pricing`
3. Now, start with chatting as `Can you provide me a summary of what this project is?`
4. First type `/terminal-setup` command for key bindings for new line, so when you are chatting and you have to go to new line then press `shift+enter` for new line in claude terminal.
5. type `can you switch to a new branch called claude-edits` , so claude will switch to a new branch because claude has knowledge about the git branches. It can run bash commands also. It can do stage and commit as well and resolve conflicts as well in git after merging. You can check changed branches in VS Code.
6. Put some github code in current local folder. Now, it's a good idea to run `\init` command initially for `claude.md` , so calude code scans entire codebase of current folder and summarize and store into claude.md file.
7. You can change the project structure in claude.md, for example, you can add a folder as `hooks` and then claude will understand it and then for your project, you can type, `can you create a hook to store user's theme preference in, when they toogle the theme on site? Store the value in local storage for next time. Don't use the hook anywhere yet.`
8. You can add memory for claude.md directly from the chat session. For example, type `# When making new page components, always add a link to that page in the header` . `#` is to memorize or you can use `/memory` command. Select `Project Memory` and check the claude.md file now.
9. You can type `Can you add a new /about page with only an h2 title and a single line of lloren as content.`
10. To select a file in claude code terminal, use `@Notes.md` and then press `Tab` key. So, it will provide the context. For example, type `Convert @Notes.md to pdf` and press enter. You can add multiple files using @ symbol in the prompt. If you select few lines of a file then it will be displayed below the claude terminal.

We can also add images as context in the claude code terminal using drag and drop of images in the claude code terminal.

11. You can use `/exit` command to exit the chat session completely. `/clear` command is used to clear the entire session context and chat history.

`/compact` command is used to compacts the chat and context into a small summary.

If you press ESC twice then it rewind to a previous point in the session.

12. If you have used `/exit` command then you can put `/resume` again after opening the claude terminal to see previous chat sessions and context and resume from those sessions/contexts.
13. Reference Material: https://code.claude.com/docs/en/overview
14. You can put `can you add a commit` for git commands.

Lecture 15 · Planning and Thinking

Claude Code can do planning and thinking. Planning and Thinking both are different things. Thinking is what claude code to do when you ask claude code to think about a solution before writing the code and Planning is to make a plan for how to add intensive features and ask you to approve it.

Type `Shift+Tab` to switch between plan mode and accept edit mode. Now, in the plan mode, you can write something as `Can you make a custom component for an Avatar (no pic, only initial) and find any places in the project where it can replace Avatar-like templates` and type Enter. You can put instructions in claude.md and put an attachement in claude code terminal.

Thinking mode is good when claude code works on more complex logic. It takes more tokens. For example, `Implement a comment system into the application, where users can authenticate and then comment on blog posts. Think hard about this implementation, including the database schema, authentication services, moderation and real-time updates.`. Here, in this example, "Think" will trigger the thinking mode and in claude code terminal, use planning mode also to know, it thinks. So, "Think" word in the prompt enables the thinking mode here. Also, Put task in chunks and then guide model for each small task.

In prompt, you can write `Think harder` or `ultra think` to think more but it will consume more tokens. Also, if after think, it gives a larger plan then accomplish the tasks in chunks or phases and don't put everything in one go.

16 · MCP/MCP Servers

MCP servers give ability to claude code to connect and communicate with external data sources, services and APIs.

MCP stands for model context protocols. It is designed by Anthropic in such a way that AI Models can interact with external data sources by providing external tools and contexts by MCP servers.

MCP servers have different tools to do different things and interact with different sources.

So, you can think MCP servers as to plugin claude code or any other client and claude is directed by an AI model to interact with the data sources for the specific tool.

For example, `Supabase MCP Server` have tools like `list_tables()`, `deploy_edge_function()`, `execute_sql()` etc. So, if I install `Supbase MCP Server` the claude code will have access to all these tools.

Another MCP server is `Playwright MCP` which provides browser automation capability using Playwright. This server enables LLMs to interact with web pages through structured accessibility snapshots, bypasssing the need for screenshots or visually-tuned models.

First run some commands on mac to add mcp servers then in claude code terminal type `/mcp` for list of mcp servers.

17 · Subagents

Each subagent can be `isolated` and configured to work on particular tasks. We can make sub-agents who can wrok as `Unit Tester` or `Security Auditor` or `Code Reviewer` or `UX Reviewer` etc. Each of these sub-agents are independent to each other and have their own system prompts and tools and context window.

Claude Code can delegate tasks to subagents. Each subagent has specialized expertise which we can fine-tune in a specific area through detailed system prompt. Since each subagent has the isolated context window, it keeps context cleaned and focused leading to better results. It also reduce context overload in the main session.

In claude code terminal, type `/agent` for list of agents. Now, create a new agent here on project level and describe what it will do.

For example, you can describe as `Expert UI and UX engineer who reviews the UI & UX of React components in a browser using Playwright, take screenshots and then offers feedback on how to improve the component in terms of visual design, user experience and accessibility.` and hit enter. Claude code will create this new agent.

You can see the newly created agent in agents folder inside .claude folder with the markdown file. You can edit this mardown file also if you want.

18. For github code and its review, you can use `/install-github-app` and then choose the current or different directory and then install and authorize claude on github. Now, create a long-lived token. Now, create and merge pull request. Now, create an issue in github repo and put comment like `@claude can you fix it` the claude will fix it.

PART 03

Tutorial 2 — Claude.ai, Connectors and Projects

1. Go to `https://claude.ai/new` for the web interface. Try various options like voice assistance.
2. You can use wispr flow AI for text-to-speech and many other applications which can save typing efforts.
3. Now, you can go to Customize on claude.ai and use the connector to create a new connector like github. There is a browse option for connectors and you can add connectors like gmail etc.

You can check the added connectors when you ask a question. Like for gmail connector, you can ask like `find important mails that need my reply urgently from my gmail`.

4. `Skills` is a set of instructions that we can give to claude for it to have specific knowledge of doing one task.
5. In web interface, we can select `web search` and ask `show me the best netflix shows that are new for me to watch in 2026`.
6. To save credits in claude so that it will not say like use after few hours.

- Don't have long chats, instead start fresh chats because in a long chat, it always takes all the previous context for a new message and it takes a lot more tokens. So, you can do as put `summarize the context` in chat and copy the context and start a new chat.
- You can edit the existing prompt if claude does not give the correct response. So, instead of giving a new prompt in later chat conversation, you can edit the existing prompt.
- Instead of asking one question at a time, combine your questions and ask them at once.
- Utilize claude projects by putting all the files in one place instead of uploading files again and again.
- Setup the permanent memory in settings. So, got `Settings` and then `Capabilities` tab and turn on the `Generate memory from chat history`.
- Use the right model. `Haiku` and `Sonnets` are good for writing and simple tasks but for deeper and long research tasks, opus is the way to go. `Opus` is a better model but also it takes more tokens/credits.
- Work in sessions like morning, afternoon, evening etc. becuase it has 5 hours window.

7. Now, In web interface, we can select `research` and ask `Research about the best mics to buy as a youtuber who is just getting started and records a lot of screen sharing tutorials from his home and has a budget of 20k INR and lives in bengaluru and wants it delivered in 3-5 days max.`. In settings, you can update the description about you in your profile so claude will know your age and your interest etc and accordingly it will give you better results. For

example, if you mention india as living place then it will give results in rupees and not in dollars. So, it means you are giving the context to claude.

8. The is an option called `Projects`. It is like a custom GPT. Create a new project called `YouTube Long form Scripts` and write the instructions as `You are a scriptwriter with over 5 years of experience writing in-depth Youtube documentary scripts for the indian audience in sectors like start-ups, technology, innovation and creativity` and create the project.

Next step is to start using it. You can upload files as context/memory to it for training and then ask a question.

9. Cowork is a new capability in claude to use the desktop and do function on it. It has access to projects and can use connectors and plugins (multiple skills at one place). Anthropic has so many plugins, so you can check it. You can install any plugin like productivity as per your requirement and then ask the question by giving access to files and folders for your work.

Using Cowork, you can schedule the task also. You can schedule the workflow automation at a particular time in a week or day etc. For example, you can schedule to extract last 24 hours best tweets and show at 10 AM daily.

You can also use Dispatch feature of cowork to connect with claude in mobile phone and give instructions to upload files and arrange them in google drive etc. via cloud in google chrome etc.

10 · Skills

It is a custom instructions that I am giving to a claude to execute a set of workflows and give me the output.

To create a new skill, just write in chat that `make a new skill`.

11. In claude code chat, you can write something `generate a snake game so that I can play in browser`.

12. You can type `claude code` in vscode terminal and then to see usage, you can use `/usage` command.

13. You can use claude in excel, powerpoint, word etc.

14. You can use claude in finance also. You can give the prompt as:

```
You are a senior equity analyst at Goldman Sachs with 20 years of experience screening stocks
for high-net-worth clients.

I need a complete stock screening framework for my investment goals.

Analyze and provide:
- Top 10 stocks matching my criteria with ticker symbols
- P/E ratio analysis compared to sector averages
- Revenue growth trends over the last 5 years
- Debt-to-equity health check for each pick
- Dividend yield and payout sustainability score
- Competitive moat rating (weak, moderate, strong)
- Bull case and bear case price targets for 12 months
```

```
- Risk rating on a scale of 1-10 with clear reasoning
- Entry price zones and stop-loss suggestions

Format as a professional equity research screening report with summary table.

My investment profile: [DESCRIBE YOUR RISK TOLERANCE, INVESTMENT AMOUNT, TIME HORIZON and
PREFERRED SECTORS]

Age: 33 years
Risk Tolerance: Medium
Investment Amount: 50 Lacs INR
Time Horizon: 5 years
Preferred Sectors: Energy, Pharma, IT, Banking
```

You can schedule in cowork with monday everyday with this prompt.

15. You can also learn using claude by giving prompt like `teach me photosynthesis visually`.

16. You can import data from chatgpt to claude using `https://claude.com/import-memory`.

PART 04

Tutorial 3 — Prompting and Extended Thinking

1. In claude chat, if you have to draft an email then you can write the prompt something like `Can you help me draft a welcome email to my new students whenever they join my English Fluency Mastery course 2.0. The email should make them feel that they have joined a premium course.`

In the generated answer, you can select the text then you can see the `Reply` option if you want a clarification on something.

2. when you are question-answering in claude code chat and after asking a question, if you like some response then it is better to save its prompt for reusability and for that you can ask that write a prompt for it.
3. For complex tasks, you can use the Extended Thinking mode.
4. If you need output in a block then write the prompt as `provide the output in a code block`.
5. For artifacts, make sure you have enabled the artifacts,AI-powered artifacts,code execution and file creation options in settings.

Some `inspiration` artifacts are available in the artifacts tab.

To create artifacts, go to chat with web search and write `Help me to generate a landing page for selling Kurkure Chips with various variety, so kindly do more research on this from the internet using web search. Write complete content with multiple pages and proper system to fill the form and getting responses.`

Once you get the answer, you will get one html file and open it side by side in the claude itself for preview and publish this artifact and generate the link so that anyone can access this artifact. For me, it is `https://claude.ai/public/artifacts/a8d2ba8b-664d-4f89-b026-f3c2dacfac8e`

6 · Claude Skills

You can go to Settings and then Capabilities and you can browse skills and create new skills also.

For example, you can create a skill by first give access claude to a skill as `slack-gif-creater` from settings--> capabilities --> browse skills --> slack-gif-creater and then going to claude chat and write:

`generate an animated gif of drinking coffee for slack`

Now, while generating response, it will show first like `Read the Slack GIF creator skill`.

Now, if you have to build a new skill, first you can see the examples by clicking on upload an skill option and there in an option below to see examples and it will navigate to `https://github.com/anthropics/skills/blob/main/skills/algorithmic-art/SKILL.md` and there are many other skills.

There are various options like create skills with claude or give skill instructions or upload zip, .skill or .md files.

`skill-creator` is also a skill that you can use to create new skills.

First we create an skill using `Create with Claude` option.

A. First write the prompt as `Let's create a skill together using your skill-creator skill. First ask me what the skill should do.`

B. Then write the prompt as `I want something where I drop my csv file and ask to create dashboard then it creates industry level dashboard with all business related ROI metrices.`

C. Then answer few questions and it will generate dashboard related files and `skill.md` file.

You can use this md file later like you can add it as prompt file and chat with claude to do more things in the dashboard.

7. You can install various plugins in the claude code desktop app like google chrome, pdf tools extension etc and go to configure and allow all permissions. After that ask like `Can you find me the best videos on the internet about how our brain works, use the chrome web connector`.
8. In claude desktop, go to settings--> developers. You will see various MCP servers there.

PART 05

Tutorial 4 — Claude Code Desktop

SOURCE: TECH WITH TIM

1. Before starting see the official documents from `https://code.claude.com/docs/en/quickstart`.
2. In Desktop version of claude code, there are 3 mode: chat, code and cowork.
3. First you have to install git on your system and so you have to go into claude code chat and ask `can u install git for me`.
4. You can ask claude to `How do I connect my github account to the terminal here?` and follow the steps to connect claude with your github account
5. Now, open Cursor IDE (it is a VSCode fork) and open a new folder `ClaudeCodeTest` and then open the terminal and type `claude`.

Now write the prompt as `Make tic tac toe that I can play on the web`.

You can run claude in multiple terminals also.

6. For any help, we can use `/help` in claude terminal.
7. In claude, there are various modes like plan mode (creates a detailed plan that claude will follow), ask mode (default mode where we ask something before the claude does something) and code mode (it's just going to code by default. It's not going to ask you for permission).

In Claude Terminal,

`accepts edits on` --> coding mode
`plan mode on` --> plan mode
`? for shortcuts` --> ask mode

Using `Shift+Tab`, you can toggle with different modes.

8. Using `ESC+ESC` means if you press escape key 2 times then you clear all the text/commands from the claude terminal. If you have to go previous text then you have to press Up Arrow key. Simliarly, if you have to go next text then you have to press Down Arrow key. And you cycle it in this way.
9. If you have to line break or go to new line then you have to press `Alt/Option + Enter/Return`.
10. While asking, you should have to clear what you want and be specific.
11. You can install `Wisper Flow` so that you don't have to type in claude terminal. You can tell and it will convert into text.
12. After getting the response from planning mode, you can use `Ctrl+O` to expand it.

13. For larger projects, you can instruct claude to create a git repo after connecting with your account in github and you can see in github also.

14. When you put `/exit` and starts the claude again then you lost all the previous history because a new session starts. So, you have to use claude.md file and memory. So, to have a persistant memory, you have to use `/init` command first and then `claude.md` will be created automatically and it is having all the information based on your codebase and the analysis.

Now, again when you put `/exit` and ask like `What do you know about this project?` then you will get the information from the claude.md file.

15. Using `/tasks` command, you can get to know about background task or any other task.

PART 06

Tutorial 5 — Plan Mode and Building a Project

1. To resume the conversation, you can use `/resume`. Supoose, you ask in the claude terminal as `hi, how are you` and then you kill the terminal and again you open the claude terminal and put the command as `/resume` to see previous conversation.
2. You can see the release notes using `/release-notes`.
3. First we create a simple project. So, first clear the context using `/clear`. Now, write in `paln mode` using `shift+tab` cas:

`I want to create a vite project with ts support and I also want to create express backend that support typescript. I want to use tanstack query for api calls. So, create a basic todo app.` click enter and kill the terminal.

Now, open another vscode terminal and install the context7 mcp server from `https://github.com/upstash/context7` and run the command as `npx --claude ctx7 setup` and then follow instructions like select MCP server and claude and login through browser. Now, in this another terminal, open claude terminal and check using `/mcp` command that mcp server is installed or not.

Now, type `claude --resume` and ask `Hey can you check latest vite docs using context7 mcp server, check how we install vite project, also check latest docs how we install tanstack query` and execute in plan mode.

When the app build is completing, you can run as:

To run it:

terminal 1

cd 6.Tutorial5/server && npm run dev # http://localhost:3001

terminal 2

cd 6.Tutorial5/client && npm run dev # http://localhost:5173

I haven't opened the browser to verify the UI renders — the backend API is verified, and both projects compile, but you'll want to load
http://localhost:5173 to confirm add/toggle/delete behave correctly visually.

open http://localhost:5173 in a browser to confirm the UI works.

4. Using `npx ccusage@latest` in vscode terminal, we can check the claude usage and cost with date.

5 · Claude Code Hooks

First copy the page info from `https://code.claude.com/docs/en/hooks` and then open a new instance of claude code terminal and then paste the page info which you copied then put `\` for new line and then ask `can you explain this hooks concept using docs` and then execute in plan mode.

Using pre-hooks, you can apply the restriction like claude will not see the `.env` file.

Now, ask `can you create a hook that prints a console log after any task is done or pending or needs for approval. Create it for local files, not global and it should be shell commands file for these hooks that you have created. Also let me know where you have created these hooks and how you can use them. Also test these hooks for me. Also create the logs as well. I am new to this so explain in depth.`

6 · Claude Subagents

Open new claude terminal instance and paste the docs from `https://code.claude.com/docs/en/sub-agents` and ask `can you explain this multi agents concept using docs. Can you create a doc for multiagent so that I can feed into next prompt. Also create a memory file claude.md in the current folder by using /init command in claude code` and then execute it.

Now, use `/exit` and then use `claude --resume` and select `sullstack-todo-app` chat and give multiagent guide file by dragging the file in claude terminal and ask `how will you use this multiagent support to do task fast. Do all the things in current folder and execute them to make app faster and improved using multiagent support.`.

Now, again use `claude --resume` after exit and reload and see agents using `/agents` command.

7. You can create slash commands like `/explain-this-file` using claude and use it in vscode terminal.

PART 07

Tutorial 6 — Coding Agents in Practice

1. First clone the repos from `https://github.com/mosh-hamedani/expense-tracker-starter` and `https://github.com/mosh-hamedani/helpdesk`. I will fix the bugs, refactor the code and add new features.
2. Claude code is an agentic coding tool. Basically, it is a coding agent. A coding agent can access to tools like write,read, bash, git etc. on itts own and do actions.

Other coding agent are github co-pilot, windsurf, cursor etc.

Claude Pricing:

https://claude.com/pricing#api
https://claude.com/pricing#team-&-enterprise
https://claude.com/pricing

3. Run the commands:
   ankit@MacBook-Air 25.Claude-Code-Deep-Dive % cd 7.Tutorial6
   ankit@MacBook-Air 7.Tutorial6 % mkdir playground
   ankit@MacBook-Air 7.Tutorial6 % cd playground
   ankit@MacBook-Air playground %
   ankit@MacBook-Air playground % git clone https://github.com/mosh-hamedani/expense-tracker-starter.git
   Cloning into 'expense-tracker-starter'...
   remote: Enumerating objects: 18, done.
   remote: Counting objects: 100% (11/11), done.
   remote: Compressing objects: 100% (11/11), done.
   remote: Total 18 (delta 0), reused 0 (delta 0), pack-reused 7 (from 1)
   Receiving objects: 100% (18/18), 29.79 KiB | 14.90 MiB/s, done.
   ankit@MacBook-Air playground % cd expense-tracker-starter
   ankit@MacBook-Air expense-tracker-starter %
4. For multiline terminal, you can type `/terminal-setup` and then using `Shift+Enter`, you can use multiline terminal. It is better than put backslash `\` after each line.
5. Now run the bash command using exclamation mark `!` at the start as `! npm run dev` in claude terminal. Output will be `sh: vite: command not found`. The better but slower way is to ask `run this app` then claude will figure out and install the dependencies after giving the permissions. Now you can use `/tasks` command to see all the background tasks.

6 · Creating Project Memory

When you start a new project then run `/init` command to generate agents.md file and also generate the claude.md file in the current folder. It has project specific architecture, prompts and other things.

7. LLMs are non-deterministic and so it can't be predictable. So, everytime, output of `/init` is different.
8. You can use git commands in bash mode like `! git add . && git commit -m "chore: update" && git push origin main` or using source control in vscode or using claude terminal like `commit` but it consume some tokens and cost but generate a beautiful commit message.
9. To make terminal elegant you can ask `make the vscode terminal and claude terminal beautiful and elegant with colors and display of git branch or amny other features so make it more beautiful and elegant with modern look like uv package manager or any other modern beautiful and elegant terminal.`

10 · Effective Prompting

- Be specific and clear

So, instead of saying "Add Authentication", say "Add JWT-Based Authentication to the login endpoint using the existing User model."

- Give Context
- Be concise

For example:

`In @src/App.jsx, total income and expenses are not calculated correctly. Fix it.`

then ask, `In transactions array, amount should be a number.`

11. we can select few lines of the code in any file and ask like `explain the selected code`.

For example:

In `App.jsx` file, select the line `date: new Date().toISOString().split('T')[0]` and then ask `explain the selected code`.

12 · Using the Plan Mode

It is useful to build new features.

For example:

Enable the plan mode and ask `add the ability to delete transactions`.

Now, when plan is made then it will ask few options like auto-accept edits, manually approve edits etc, So, select `Tell Claude what to change` and then put `add the confirmation dialogue box` in the plan mode itself for this option. Now, select auto-accept edits option.

13 · Managing the Context

Think Context Window as claude's working memory. So, it is like claude can see at any given moment. Context Window can contain conversations, files, claude.md etc.

Now, put the `/context` command and see the content. Roughly Token is 3/4th of a word.

System tools are like bash command tools, read, write, git etc related tools.

Now, we have `/compact` command which we use when context window gets full and we are working on a related task and we don't want to lose history or context. Claude will put the summary in context and clear the original context after the execution of this command.

So, after executing `/compact` and then `/context`, we can see less consumption of tokens as compared to previous output of `/context`.

`/clear` is used to clear the context and conversation history. So, to clear unrelated task, use it.

Now, again if we use `/clear` and then `/context`, we get different output. (Why?)

14. Claude automatically compacts at ~70%.

15 · Tracking cost and usage

`/cost` for API usage and `/usage` for Claude subscription plan.

16 · Cost Management

Use

- /clear
- /compact
- Keep Claude.md file lean and relevant because this file goes to model for each request.